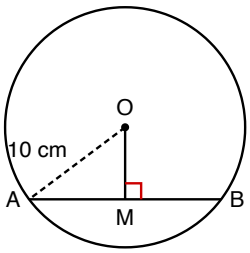
Shape and Space 6 — Chords · Alternate segment theorem · Intersecting chords

Calculator allowed. Marks in brackets. ★ = harder question. Diagrams are not drawn accurately. Give a reason for every angle fact you use.

1 O is the centre of a circle of radius 10 cm. AB is a chord of length 16 cm and M is the mid-point of AB. [3]

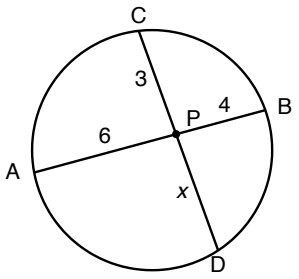


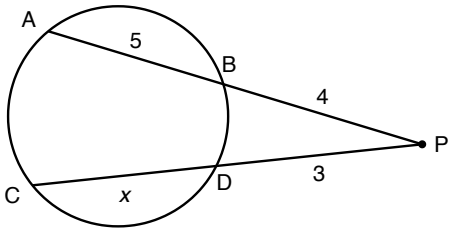
a Write down the length of OA.

b Write down the length of AM.

c Work out the length of OM.

2 In each diagram find x. The lengths are in cm. [6]

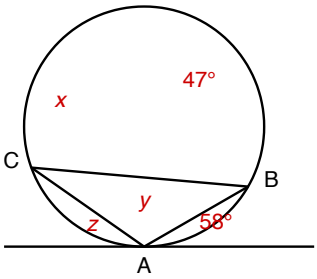


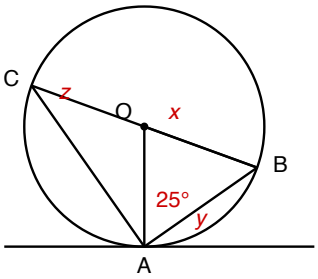


a The chords AB and CD cross at P inside the circle. AP $= 6$, PB $= 4$, CP $= 3$, PD $= x$.

b P is outside the circle. PB $= 4$, BA $= 5$, PD $= 3$, DC $= x$.

3 In each diagram find the angles x, y and z. Give a reason for each step. [12]

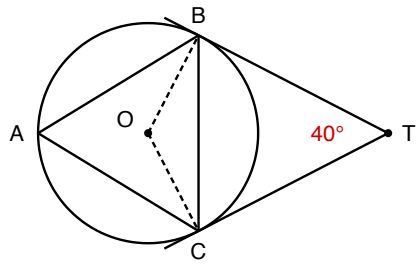




a The line through A is a tangent. The angle between the tangent and the chord AB is 58°, and angle ABC $= 47^\circ$.

b O is the centre and the line through A is a tangent. Angle OAB $= 25^\circ$.

4 TB and TC are tangents from T to the circle with centre O. A is a point on the major arc BC. Angle BTC $= 40^\circ$. [4]



a Find angle TBC. **b** Find angle BAC. **c** Find angle BOC.

Total: 25 marks

Check: perpendicular from the centre bisects the chord (then Pythagoras) · alternate segment: tangent–chord angle = angle in the opposite segment · intersecting chords, inside or outside: $\mathrm{AP} \times \mathrm{PB} = \mathrm{CP} \times \mathrm{PD}$ with every length measured from the crossing point P · tangent $\perp$ radius · two tangents from a point are equal.